

WASP-12b

R-2155

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150115 · created 2026-08-12T23:03:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457037.79718 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1026 +/- 0.0088
Transit depth (Rp/Rs)^2	1.05 +/- 0.18 [%]
Orbital Inclination (inc)	84.06 +/- 1.96
Airmass coefficient 1 (a1)	1.1062 +/- 0.0074
Airmass coefficient 2 (a2)	-0.085 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	None
Optimal Aperture	4.23
Optimal Annulus	6.59
Transit Duration (day)	0.1234 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1026 ± 0.0088 vs 0.1178 (deviation 1.73σ)
Duration fit vs published	0.1234 d vs 0.125 d (deviation 0.46σ)
Mid-time O–C	-0.3 min
Depth SNR	11.7
Residual scatter	0.67 %

Light curve

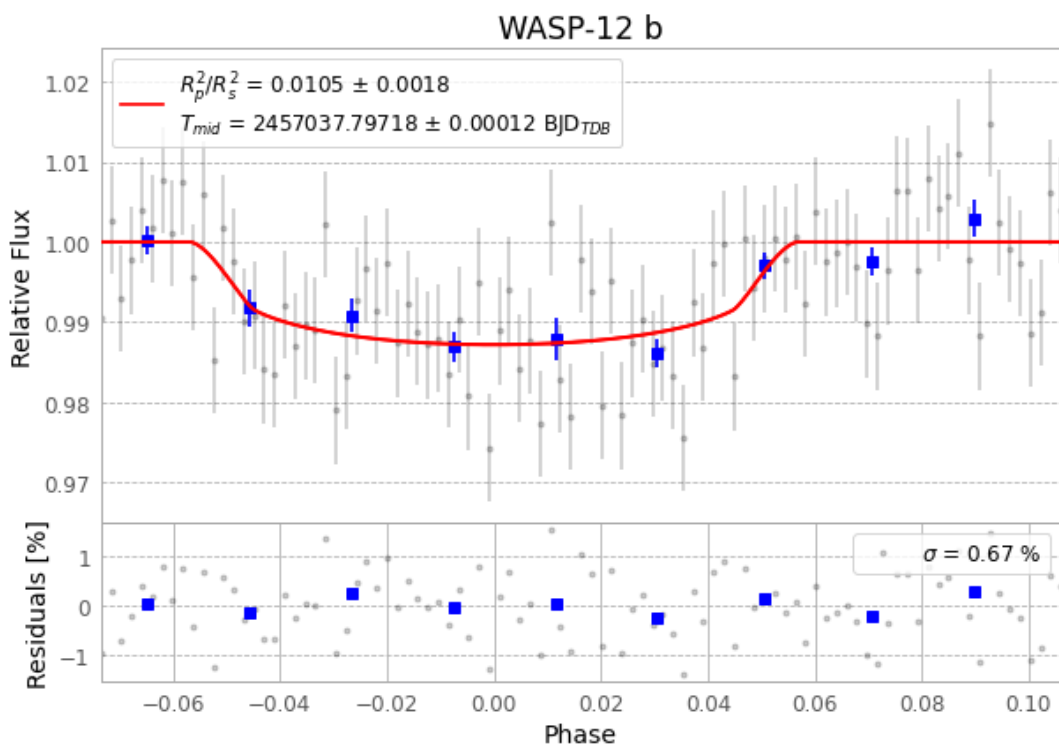


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-14.png (SHA-256 a4298d5293425194...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96752df6c1c82afa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

95 FITS frames (62 MB), WASP-12150115050312.FITS → WASP-12150115094512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150115.log (ed6050006a76...), FinalLightCurve_WASP-12 b_2015-01-14.png (a4298d529342...), FinalLightCurve_WASP-12 b_2015-01-14.csv (d8cc667b90af...)

Compute resources

Wall time 128.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:03Z.